IMMERSIONS, CLOSED AND OPEN IMMERSIONS ARE CLOSED UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5084-5114, following the statement at 5077-5082.

Allowed standard prerequisites (editorial).

- Exactness of pushforward along closed immersions. Modules, Lemma modules-lemma-i-star-exact (cross-chapter, cited at ~5097). b_* is exact, giving surjectivity of b_*O_Y -> b_*a_*O_Z.
- Characterization of closed immersions. lemma-characterize-closed-immersions (same chapter, cited at ~5098). Homeomorphism onto a closed subset + surjective structure map => closed immersion.

Source reference. Original lemma 01LD: sources/stacks-schemes/pinned-original-source.tex, source lines 5044–5044. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01AX: focused-seeds/geometry/S9-schemes-composition-immersion/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *A composition of immersions of schemes is an immersion, a composition of closed immersions of schemes is a closed immersion, and a composition of open immersions of schemes is an open immersion.*

Proof. This is clear for the case of open immersions since an open subspace of an open subspace is also an open subspace.

Suppose $a : Z \to Y$ and $b : Y \to X$ are closed immersions of schemes. We will verify that $c = b \circ a$ is also a closed immersion. The assumption implies that a and b are homeomorphisms onto closed subsets, and hence also $c = b \circ a$ is a homeomorphism onto a closed subset. Moreover, the map $\mathcal{O}_X \to c_*\mathcal{O}_Z$ is surjective since it factors as the composition of the surjective maps $\mathcal{O}_X \to b_*\mathcal{O}_Y$ and $b_*\mathcal{O}_Y \to b_*a_*\mathcal{O}_Z$ (surjective as b_* is exact, see Modules, Lemma 01AX). Hence by Lemma 01LD above c is a closed immersion.

Finally, we come to the case of immersions. Suppose $a : Z \to Y$ and $b : Y \to X$ are immersions of schemes. This means there exist open subschemes $V \subset Y$ and $U \subset X$ such that $a(Z) \subset V$, $b(Y) \subset U$ and $a : Z \to V$ and $b : Y \to U$ are closed

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immersions. Since the topology on Y is induced from the topology on U we can find an open $U' \subset U$ such that $V = b^{-1}(U')$. Then we see that $Z \to V = b^{-1}(U') \to U'$ is a composition of closed immersions and hence a closed immersion. This proves that $Z \to X$ is an immersion and we win. $\square$